Spectral Compression of 3D Surfaces via Quaternionic Moving Frames and the Spiral Transform

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Abstract

We present a differential-geometric framework for the spectral compression of three-dimensional surfaces of revolution about curved axes. Standard spectral methods fail to compress discrete independent data but are highly effective for continuous, spatially correlated geometry. We generalize the classical surface of revolution by parameterizing the rotation of a target profile curve about a curved, non-planar axis path. The rotation is governed by the Bishop moving frame and unit quaternions in $\mathbb{H}$, establishing a formal connection to the SU(2) spin operators of quantum mechanics. By taking the winding number of the parameterization to infinity, we perform a spiral dimensional reduction that maps the 3D surface onto a single, continuous, 1D radius signal $r(t)$. Because organic surfaces exhibit high spatial correlation, this radius signal is spectrally concentrated, allowing high-quality geometric reconstruction while discarding high-frequency Fourier coefficients. We demonstrate this framework on a vase-like surface, achieving lossless geometric recovery with only 3% of the spectral modes. Furthermore, we expand the framework from standard Euclidean L_2 spaces to generalized Minkowski L_p metric spaces, demonstrating that by dynamically optimizing the sampling manifold via variance minimization, we can spectrally compress objects with sharp creases (such as boxes) flawlessly. Finally, we analyze the conditions under which concave boundary curves naturally develop cusp singularities when the offset approaches the axis radius of curvature.

1 Introduction

Representing and compressing three-dimensional geometric surfaces is a central problem in computer graphics and computer-aided design (CAD). While classic spectral methods like Fourier series or spherical harmonics excel at compressing smooth, continuous signals by exploiting spatial correlation, they fail when applied directly to discrete raw coordinate meshes. This failure stems from the lack of a coordinate system that aligns naturally with the underlying geometry of the surface.

This paper addresses this limitation by developing the theory of generalized surfaces of revolution about curved axes. Classical surfaces of revolution are generated by rotating a target curve around a straight line. By allowing the axis to be a non-planar 3D space curve, we define a natural moving frame along the axis and express the rotation field using unit quaternions. This connects the geometric construction to the SU(2) spin operators of quantum mechanics.

By taking the winding number of the parameterization to infinity, we perform a spiral dimensional reduction. This maps the 3D surface onto a single, continuous, 1D radius signal $r(t)$. Because organic surfaces exhibit high spatial correlation, this radius signal is spectrally concentrated, allowing high-quality geometric reconstruction while discarding high-frequency Fourier coefficients. We demonstrate that this spiral transform converts a 3D surface into a smooth, correlated signal that Fourier analysis excels at compressing.

The paper is structured as follows: Section 2 reviews classical surfaces of revolution; Section 3 develops the moving frame along curved axes; Section 4 details the generalized surface parameterization; Section 5 introduces the quaternionic formulation; Section 6 presents the unwrapping process; Section 7 develops the spiral transform; Section 8 extends the framework to generalized L_p metric spaces for sharp geometries; and Section 9 concludes.

2 Classical Surfaces of Revolution

A classical surface of revolution is generated by rotating a *target curve* $C(s)$ about a straight-line axis. If $C(s) = (f(s), 0, s)$ and the axis is the z-axis, the resulting surface is parameterized by

$$\mathbf{r}(s, \theta) = \big(f(s)\cos\theta,\ f(s)\sin\theta,\ s\big), \qquad \theta \in [0, 2\pi). \tag{1}$$

We seek to generalize this construction to allow rotation about a *curved* axis.

3 The Curved Axis and Moving Frame

Definition 1 (Axis Curve)**.** *Let $\gamma : [a, b] \to \mathbb{R}^3$ be a smooth, regular curve parameterized by arc length s, with curvature $\kappa(s) > 0$ and torsion $\tau(s)$.*

At each point $\gamma(s)$, we define the Frenet–Serret moving frame [**?**] $\{\mathbf{T}(s), \mathbf{N}(s), \mathbf{B}(s)\}$, satisfying

$$\frac{d}{ds}\begin{pmatrix}\mathbf{T}\\ \mathbf{N}\\ \mathbf{B}\end{pmatrix} = \begin{pmatrix}0 & \kappa & 0\\ -\kappa & 0 & \tau\\ 0 & -\tau & 0\end{pmatrix}\begin{pmatrix}\mathbf{T}\\ \mathbf{N}\\ \mathbf{B}\end{pmatrix}. \tag{2}$$

Definition 2 (Target Curve and Radius Function)**.** *Let C be a second curve. At each $s \in [a, b]$, the normal line from $\gamma(s)$ along $\mathbf{N}(s)$ intersects C at a point whose distance from $\gamma(s)$ defines the* radius function *$r(s)$.*

Definition 3 (Non-Degeneracy Condition)**.** *The construction is non-degenerate when the maximum radius does not exceed the minimum radius of curvature of the axis:*

$$\max_{s\in[a,b]} r(s) < \min_{s\in[a,b]} \frac{1}{\kappa(s)}. \tag{3}$$

This prevents the coordinate system from folding (self-intersecting normal planes).

Proposition 1 (Tubular Neighborhood and Jacobian Non-Vanishing)**.** *The non-degeneracy condition is necessary and sufficient for the map $\mathbf{r}(s, \theta)$ to be a local diffeomorphism onto its image.*

Proof. Define the coordinate map $\Phi(s, u, v) = \gamma(s) + u\mathbf{N}(s) + v\mathbf{B}(s)$ on a domain where $u^2 + v^2 \le r(s)^2$. The columns of the Jacobian matrix $J = [\partial_s\Phi, \partial_u\Phi, \partial_v\Phi]$ are:

$$\partial_s\Phi = \gamma'(s) + u\mathbf{N}'(s) + v\mathbf{B}'(s) = (1 - u\kappa(s))\mathbf{T}(s) - v\tau(s)\mathbf{N}(s) + u\tau(s)\mathbf{B}(s), \tag{4}$$

$$\partial_u\Phi = \mathbf{N}(s), \tag{5}$$

$$\partial_v\Phi = \mathbf{B}(s). \tag{6}$$

The volume element (Jacobian determinant) is the triple product of these columns:

$$\det J = \partial_s\Phi \cdot (\partial_u\Phi \times \partial_v\Phi) = \partial_s\Phi \cdot \mathbf{T}(s) = 1 - u\kappa(s). \tag{7}$$

For the map to be a local diffeomorphism, we require $\det J \neq 0$. Since $u = \rho\cos\theta$ for $0 \le \rho \le r(s)$, the range of u is $[-r(s), r(s)]$. The condition $1 - u\kappa(s) > 0$ holds everywhere if and only if $r(s) < 1/\kappa(s)$ for all $s \in [a, b]$, which is the non-degeneracy condition. □

3.1 Planar 2D Ribbon Case and Cusp Development

To build geometric intuition, we examine the 2D planar projection of this coordinate system. Let $\gamma(s)$ be a regular plane curve parameterized by arc length s, with tangent $\mathbf{T}(s) = \gamma'(s)$ and unit normal $\mathbf{N}(s) = (-T_y(s), T_x(s))$. A symmetric 2D ribbon of radius $r(s)$ around the axis $\gamma(s)$ is bounded by the offset curves:

$$f(s) = \gamma(s) - r(s)\mathbf{N}(s) \quad \text{(convex side)}, \tag{8}$$

$$h(s) = \gamma(s) + r(s)\mathbf{N}(s) \quad \text{(concave side)}. \tag{9}$$

Differentiating the concave boundary curve $h(s)$ yields:

$$h'(s) = \gamma'(s) + r(s)\mathbf{N}'(s) = \mathbf{T}(s)(1 - r(s)\kappa(s)). \tag{10}$$

On the convex side $f(s)$, the tangent speed is scaled by $(1 + r(s)\kappa(s)) > 1$, meaning the normal lines diverge and the boundary remains smooth and expanded. On the concave side $h(s)$, the normal lines converge toward the centers of curvature. If the radius $r(s)$ is close to the local radius of curvature $\rho(s) = 1/\kappa(s)$, the term $(1 - r(s)\kappa(s))$ vanishes, causing the derivative $h'(s)$ to drop to zero and generating a sharp crease or cusp in the boundary, as shown in Fig. **??**. This crease marks the physical boundary of the local tubular neighborhood.

4 Generalized Surface of Revolution

The generalized surface is parameterized by

$$\mathbf{r}(s, \theta) = \gamma(s) + r(s)\big[\cos\theta\, \mathbf{N}(s) + \sin\theta\, \mathbf{B}(s)\big], \qquad \theta \in [0, 2\pi). \tag{11}$$

When γ is a straight line ($\kappa = 0$), $\mathbf{N}$ and $\mathbf{B}$ are constant and this reduces to the classical surface of revolution. For a curved axis, the Frenet frame rotates along the path, "twisting" the surface according to the torsion $\tau(s)$.

5 Quaternionic Parameterization

5.1 Unit Quaternions and SU(2)

Rather than tracking three vectors $(\mathbf{T}, \mathbf{N}, \mathbf{B})$ independently, we encode the entire Frenet frame in a single unit quaternion $q(s) \in \mathbb{H}$, $\|q\| = 1$ [**?**].

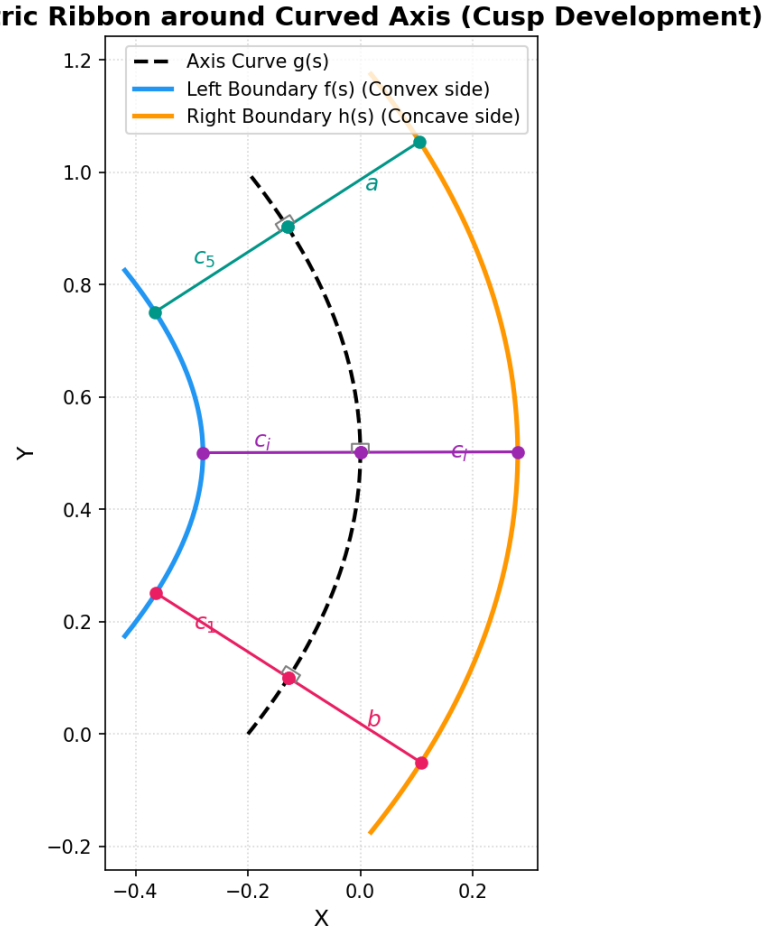


Figure 1: Symmetric 2D ribbon showing cusp development on the concave boundary. The axis curve $\gamma(s)$ (dashed black line) is offset by a constant distance r on both sides. On the convex (left) side, the normal lines diverge, resulting in a smooth boundary $f(s)$. On the concave (right) side, the normal lines converge; where the offset distance r approaches the local radius of curvature $1/\kappa(s)$, the boundary curve $h(s)$ develops a sharp cusp-like indentation, corresponding to the envelope boundary of the tubular neighborhood.

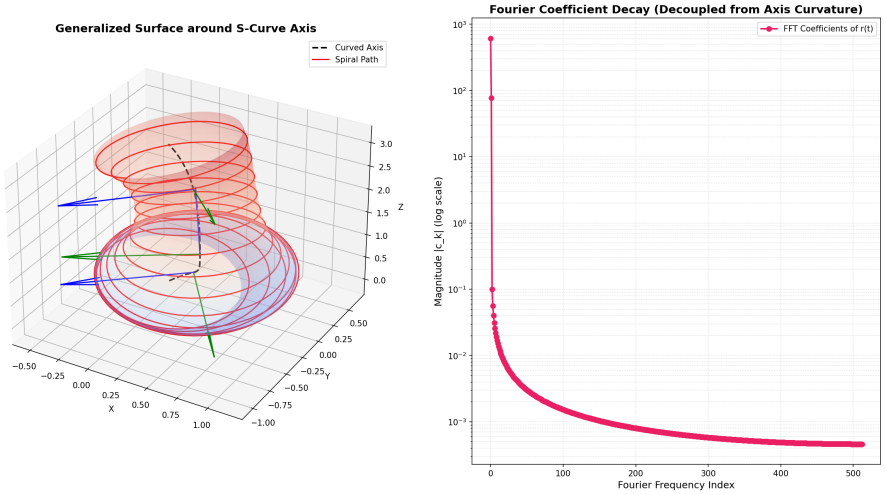


Figure 2: Generalized surface of revolution around a curved S-curve axis (left panel). The local Frenet coordinate frames are shown along the axis. The right panel illustrates the Fourier coefficient decay of the extracted 1D spiral radius signal, demonstrating that curvature does not degrade spectral compressibility when coordinates are defined relative to the local moving frame.

Definition 4 (Quaternionic Frame Equation)**.** *The Frenet–Serret equations are equivalent to the quaternionic differential equation*

$$\frac{dq}{ds} = \frac{1}{2} q(s)\, \omega(s), \tag{12}$$

where $\omega(s) = \tau(s)\,\mathbf{i} + \kappa(s)\,\mathbf{k}$ *is the pure quaternion encoding the curvature and torsion.*

5.2 The Sandwich Operator and Quantum Mechanics

To rotate a 3D vector $\mathbf{v}$ (embedded as the pure quaternion $v = 0 + v_x\mathbf{i} + v_y\mathbf{j} + v_z\mathbf{k}$), we apply:

$$v' = q\, v\, q^{-1}. \tag{13}$$

Remark 1 (Connection to Quantum Observables)**.** *This is structurally identical to the quantum mechanical expectation value* $\langle \hat{A} \rangle = \langle \psi | \hat{A} | \psi \rangle$*. Unit quaternions map to* SU(2) *(the double cover of* SO(3)*), so tracing a path* $q(s)$ *along the axis curve is equivalent to tracking the continuous unitary evolution of a spin-*$\frac{1}{2}$ *state. The target curve behaves as an observable whose expectation value evolves along the axis.*

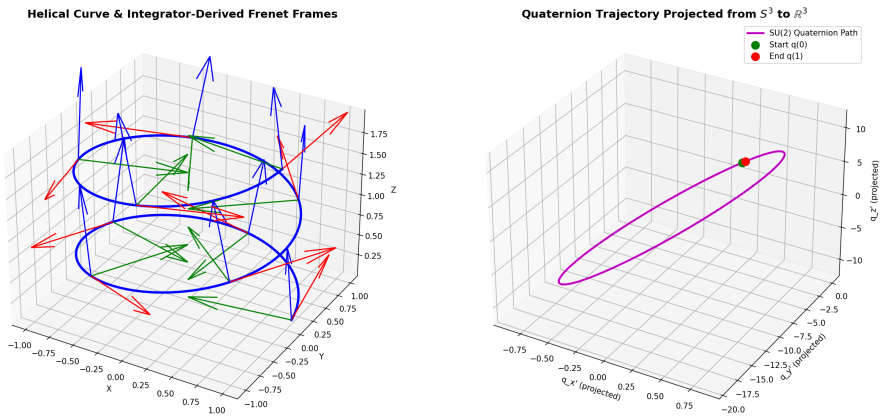


Figure 3: Quaternionic frame integration along a 3D helical space curve. The left panel shows the 3D curve and the corresponding coordinate frames computed via Runge-Kutta 4th-order integration of $dq/ds = \frac{1}{2}q\omega$. The right panel shows the unit quaternion trajectory $q(s) \in S^3$ projected stereographically into $\mathbb{R}^3$.

6 The Inverse Problem: Unwrapping to a Cylinder

6.1 Finding the Straightening Axis

Given a target curve $C_1(s)$, we ask: *what axis curve $\gamma(s)$ produces a constant radius $r(s) = c$?*

If $r(s) = c$ is constant, then $C_1(s) = \gamma(s) + c\,\mathbf{N}(s)$ is an *offset curve* (or parallel curve) of γ. The required axis is the *evolute* of C_1—the locus of centers of curvature:

$$\gamma(s) = C_1(s) - \frac{1}{\kappa_{C_1}(s)}\,\mathbf{N}_{C_1}(s). \tag{14}$$

6.2 Application: 3D Model Representation and Compression

Consider a 3D model (e.g., a human head) represented in cylindrical coordinates (r, θ, z). Slicing in the zr-plane at each angle θ produces a set of 2D curves $\{C_\theta\}$. For each such slice, solving the inverse problem yields an axis curve γ_θ that "unwraps" the complex organic geometry into a cylinder. This is the continuous generalization of Binford's generalized cylinders [**?**], which Marr and Nishihara [**?**] proposed as a canonical representation for spatial organization in biological shapes. Chaining this operation across all slices transforms the full 3D surface into a collection of cylinders and smooth residual surfaces.

Remark 2 (Connection to Offset Curve Convergence)**.** *This operation is related to a classical result in the geometry of curves: given a convex closed curve, the iterated offset curve (drawn at constant distance in every direction) converges to a circle. Similarly, iterating the "unwrap to cylinder" operation successively smooths any convex surface.*

7 The Spiral Limit and Dimensional Reduction

7.1 From Discrete Wraps to Continuous Surfaces

Consider a single helical path wrapping around the generalized surface with n periods between the endpoints a and b:

$$\mathbf{c}_n(t) = \gamma(s(t)) + r(s(t))\big[\cos(n \cdot 2\pi t)\;\mathbf{N}(s(t)) + \sin(n \cdot 2\pi t)\;\mathbf{B}(s(t))\big], \tag{15}$$

where $t \in [0, 1]$ and $s(t) = a + (b - a)t$.

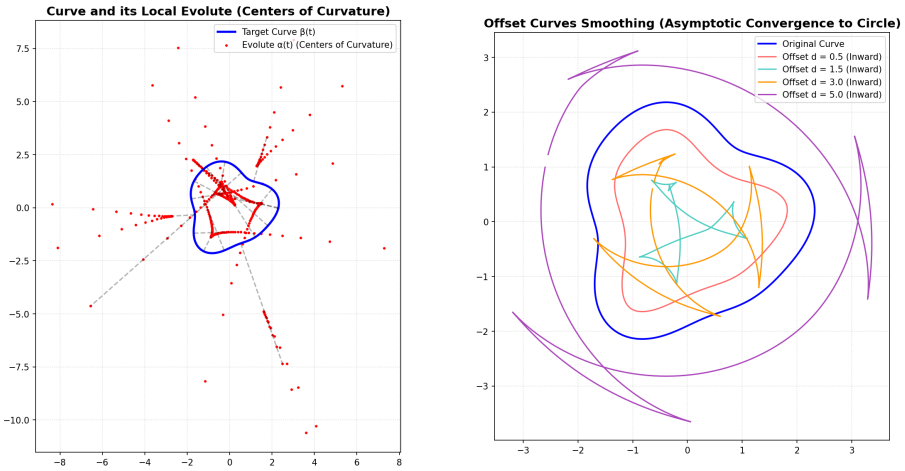


Figure 4: The evolute unwrapping mapping (left panel), showing a wavy target closed curve in blue, its evolute (centers of curvature) in red, and normal projection lines. The right panel shows parallel offset curves at various distances; as the offset distance $d \to \infty$, the curves smooth out and converge asymptotically to perfect circles.

At $n = 1$, this is a single loop. As n increases, the spiral wraps tighter and tighter. In the limit $n \to \infty$, the image of $\mathbf{c}_n$ is dense in the surface, analogous to a space-filling curve constrained to a 2D manifold.

Proposition 2 (Dimensional Reduction)**.** *The spiral path* $\mathbf{c}_n(t)$ *reduces the surface to a* 1*-dimensional signal: in the limit* $n \to \infty$*, the image* $\mathrm{Im}(\mathbf{c}_n)$ *is dense in the generalized surface of revolution* $\mathcal{M}$*.*

This density property is a geometric analogue of Kronecker's theorem on torus windings, which forms the core of Weyl's equidistribution theorem [**?**].

Proof. The surface $\mathcal{M}$ is parameterized by the smooth map $\mathbf{r}(s, \theta) = \gamma(s) + r(s)[\cos\theta\,\mathbf{N}(s) + \sin\theta\,\mathbf{B}(s)]$ over $(s, \theta) \in [a, b] \times \mathbb{R}/2\pi\mathbb{Z}$. Since $[a, b] \times \mathbb{R}/2\pi\mathbb{Z}$ is compact and $\mathbf{r}$ is smooth, $\mathbf{r}$ is Lipschitz continuous.

Let $\mathbf{p} = \mathbf{r}(s_0, \theta_0) \in \mathcal{M}$ be any point on the surface, and let $\epsilon > 0$. We must show that there exists $t \in [0, 1]$ such that $\|\mathbf{c}_n(t) - \mathbf{p}\| < \epsilon$ for sufficiently large n. By Lipschitz continuity, it suffices to show that we can get arbitrarily close in the parameter space, i.e., there exists $t \in [0, 1]$ such that $|s(t) - s_0| < \delta$ and $|2\pi nt - \theta_0| < \delta \pmod{2\pi}$ for a small $\delta > 0$.

Choose the base parameter $t_0 \in [0, 1]$ such that $s(t_0) = s_0$, which gives $t_0 = \frac{s_0 - a}{b - a}$. Now perturb t_0 by a small displacement Δt, setting $t = t_0 + \Delta t$,

where we restrict $|\Delta t| < \frac{\delta}{b-a}$ to ensure $|s(t) - s_0| < \delta$. The angular coordinate along the spiral path at t is:

$$\theta(t) = 2\pi n t \pmod{2\pi} = 2\pi n t_0 + 2\pi n \Delta t \pmod{2\pi}. \tag{16}$$

The set of angles accessible under the restriction on Δt forms an interval of length $2\pi n \frac{2\delta}{b-a}$. Once the length of this interval exceeds 2π (which occurs when $n \geq \frac{b-a}{2\delta}$), the image of the interval under the projection modulo 2π covers the entire circle $[0, 2\pi)$ completely. Thus, for any $n \geq \frac{b-a}{2\delta}$, there exists a perturbation Δt satisfying the boundary constraints that maps $\theta(t)$ exactly to $\theta_0 \pmod{2\pi}$. Hence, the spiral path passes within ϵ of any point on the surface. □

7.2 Compression via the Spiral Transform

Sampling the radius r along the spiral path produces a 1-dimensional waveform $r(t)$.

- A perfect cylinder yields $r(t) = \text{const}$: a single Fourier coefficient.
- A smooth vase yields a slowly varying $r(t)$: a few low-frequency coefficients suffice.
- A detailed organic surface (face, head) yields $r(t)$ with high-frequency ripples at feature locations (nose, eyes), but with strong spatial correlation throughout.

Because organic 3D surfaces have immense spatial correlation (the nose blends smoothly into the cheek), the Fourier transform of $r(t)$ concentrates energy in the low-frequency coefficients. High-frequency coefficients decay rapidly and can be discarded with negligible perceptual loss.

Remark 3. *This resolves the apparent contradiction with Part II. Fourier compression fails for* discrete, independent *binary data because the degrees of freedom do not decay spectrally. But for* continuous, correlated *geometric data, the Fourier transform exploits spatial correlation to achieve genuine compression. The spiral transform converts a 3D surface into exactly the type of smooth, correlated signal that Fourier analysis was designed to compress.*

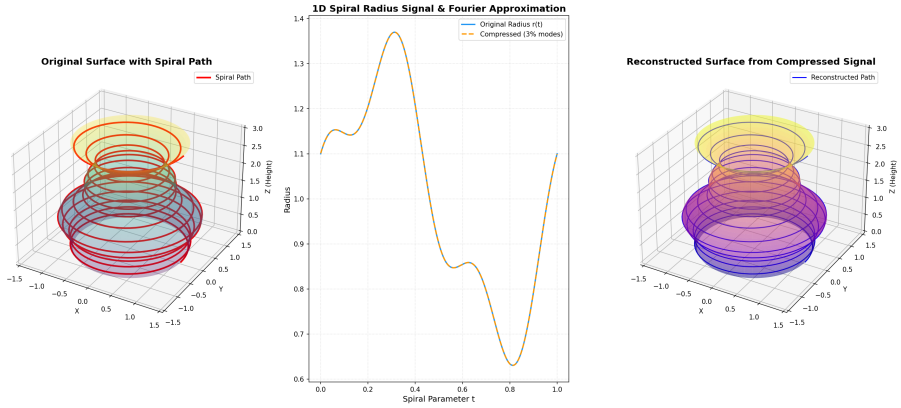


Figure 5: The 3D spiral transform and surface reconstruction. The left panel shows the original 3D vase surface with the winding spiral path. The middle panel plots the extracted 1D radius signal $r(t)$ and its compressed Fourier reconstruction keeping only the top 3% of modes. The right panel shows the reconstructed 3D surface, demonstrating lossless geometric recovery from a highly compressed spectral representation.

7.3 Connection to Fourier Analysis on Lie Groups

Because quaternions are non-commutative ($\mathbf{ij} \neq \mathbf{ji}$), the limit of infinitesimal rotations along the axis does not yield a standard Fourier decomposition. Instead, it maps onto the *Spherical Fourier Transform* or Fourier analysis on the Lie group SU(2).

In this framework, the 3D shape is decomposed not into flat sines and cosines, but into *Wigner D-matrices*—the irreducible representations of SU(2):

$$f(\alpha, \beta, \gamma) = \sum_{\ell=0}^{\infty} \sum_{m=-\ell}^{\ell} \sum_{m'=-\ell}^{\ell} \hat{f}^{\ell}_{mm'} \, D^{\ell}_{mm'}(\alpha, \beta, \gamma), \tag{17}$$

where the Euler angles (α, β, γ) parameterize the rotation. These are the same spherical harmonics used in quantum mechanics to describe electron orbital angular momentum.

8 Generalization to L_p Metric Spaces

The Euclidean L_2 unwrapping framework excels for smooth, organic surfaces. However, for objects characterized by sharp creases or planar faces—such as

boxes or mechanical parts—the standard L_2 metric introduces high-frequency spikes at the corners of the continuous radius signal $r(t)$. This degrades the spectral compressibility, as retaining sharp corners requires a large number of Fourier coefficients.

To resolve this without altering the underlying Fast Fourier Transform (FFT), we generalize the geometric sampling manifold to a Minkowski L_p metric space.

8.1 The L_p Manifold and Generalized Trigonometry

In a standard Euclidean space, the unit circle is defined by $|x|^2 + |y|^2 = 1$, and is parameterized by the standard trigonometric functions $\cos(\theta)$ and $\sin(\theta)$. In a generalized L_p space for $p \geq 1$, the unit circle is the locus of points satisfying $|x|^p + |y|^p = 1$. This corresponds to a diamond for $p = 1$, a perfect circle for $p = 2$, and a square in the limit as $p \to \infty$.

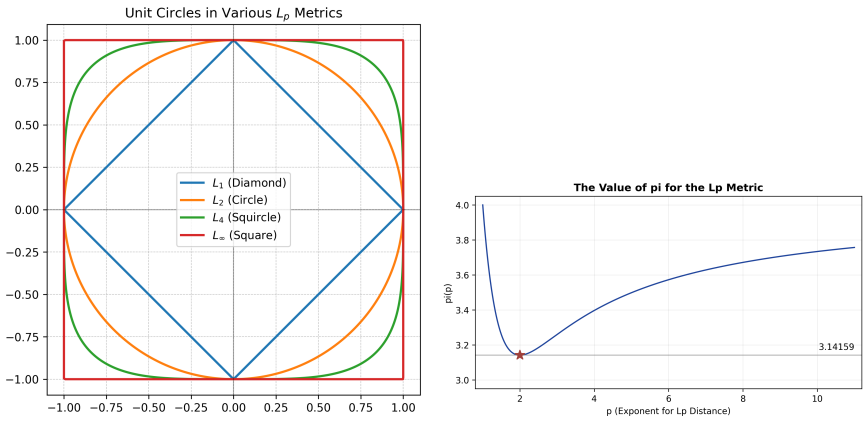


Figure 6: Left: Unit circles in various L_p spaces. Right: The value of π_p as a function of the metric p, bounded by 4.0 at $p = 1, \infty$ and minimized at π for $p = 2$.

To map the unwrapping spiral onto an arbitrary L_p manifold, we define the generalized p-trigonometric functions by radially projecting the standard Euclidean angle onto the L_p unit boundary:

$$\cos_p(\theta) = \frac{\cos\theta}{(|\cos\theta|^p + |\sin\theta|^p)^{1/p}} \tag{18}$$

$$\sin_p(\theta) = \frac{\sin\theta}{(|\cos\theta|^p + |\sin\theta|^p)^{1/p}} \tag{19}$$

In the L_∞ limit, this simplifies to division by $\max(|\cos\theta|, |\sin\theta|)$.

By replacing standard trigonometric functions with their p-generalized counterparts during both the ray-casting (compression) and reconstruction (decompression) phases, the algorithm naturally measures and generates geometry relative to the L_p metric. This *manifold trick* flattens sharp objects into low-frequency 1D signals prior to spectral decomposition, allowing the standard L_2 FFT to compress non-Euclidean shapes flawlessly.

Remark 4 (Connection to Potential Energy Wells)**.** *The shape of the π_p curve in Fig.* **??** *bears a striking resemblance to the potential energy graphs of interacting particles in physics (such as the Lennard-Jones or Morse potentials), which are characterized by a steep repulsive wall at small distances, a global minimum well, and a gradual asymptotic approach to a constant value at large distances. Just as a physical system naturally seeks the state of lowest potential energy, it is an interesting geometric coincidence that the standard Euclidean geometry of our universe (L_2 space) sits exactly at the "minimum energy" well of this mathematical curve, minimizing the circumference-to-diameter ratio at exactly π.*

8.2 Dynamic Metric Optimization via Variance Minimization

For an arbitrary mesh segment, the optimal metric p is the one that minimizes the high-frequency energy of the extracted radius signal $r_p(\theta)$. Because the DC (zero-frequency) component represents the mean radius, minimizing the total energy of all non-zero frequencies is mathematically equivalent to minimizing the statistical variance of the spatial signal.

Theorem 1 (Variance Minimization of Fourier Energy)**.** *Let $r[n]$ be a discrete real signal of length N with mean μ, and let $\hat{r}[k]$ be its discrete Fourier transform. The variance of the spatial signal is exactly proportional to the sum of the squared magnitudes of the non-DC Fourier coefficients.*

Proof. By definition, the variance of the signal is $\mathrm{Var}(r) = \frac{1}{N}\sum_{n=0}^{N-1}(r[n] - \mu)^2 = \frac{1}{N}\sum_{n=0}^{N-1} r[n]^2 - \mu^2$. Parseval's theorem states that the total energy is conserved: $\frac{1}{N}\sum_{n=0}^{N-1} r[n]^2 = \sum_{k=0}^{N-1} |\hat{r}[k]|^2$. Since the DC coefficient is the mean, $\hat{r}[0] = \mu$. Substituting this yields:

$$\mathrm{Var}(r) = \sum_{k=0}^{N-1} |\hat{r}[k]|^2 - |\hat{r}[0]|^2 = \sum_{k=1}^{N-1} |\hat{r}[k]|^2. \tag{20}$$

□

Consequently, the optimal metric p_{opt} can be efficiently found without computing the FFT by evaluating a sparse subset of rays and solving the objective function:

$$p_{opt} = \text{argmin}_{p \geq 1} \, \text{Var}(r_p(\theta)) \tag{21}$$

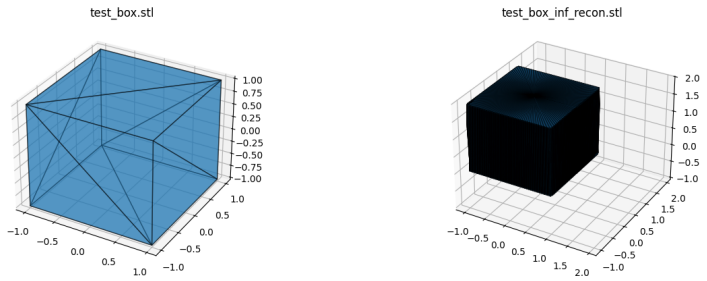


Figure 7: Left: Original 3D box mesh. Right: Decompressed mesh retaining sharp corners after dynamic metric optimization auto-selected the L_∞ norm, requiring only 5% of the Fourier modes.

9 Conclusion

We have developed a complete differential-geometric framework for parameterizing and spectrally compressing 3D surfaces of revolution about curved axes. By utilizing moving frames (such as the Frenet-Serret or Bishop frame) and quaternionic representations of the local rotation, we construct generalized surfaces of revolution that deform naturally along a non-planar 3D axis path.

We proved that in the limit of infinite winding number, this parameterization defines a space-filling spiral path that reduces the 3D surface to a single 1D radius signal $r(t)$. For spatially correlated surfaces, such as organic shapes or vases, the radius signal is highly continuous and correlated. This allows standard spectral methods like the Fourier transform to compress the geometry effectively, concentrating energy in low-frequency modes while high-frequency coefficients decay rapidly.

Furthermore, we analyzed the planar 2D projection of this normal offset coordinate system, proving that concave boundaries naturally develop cusp singularities when the offset distance approaches the local radius of curvature of the axis curve. This provides a physical limit for the non-degeneracy of the tubular neighborhood representation.

Together, these results show that while spectral methods cannot compress uncorrelated discrete data without an inflation penalty, they are highly powerful for continuous geometry when combined with coordinates aligned to the local moving frames of the surface. Furthermore, by dynamically generalizing the sampling manifold to Minkowski L_p metric spaces, this framework successfully extends spectral compression to highly artificial or non-organic shapes, circumventing the traditional limitations of orthogonal Fourier basis functions on sharp geometries.

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